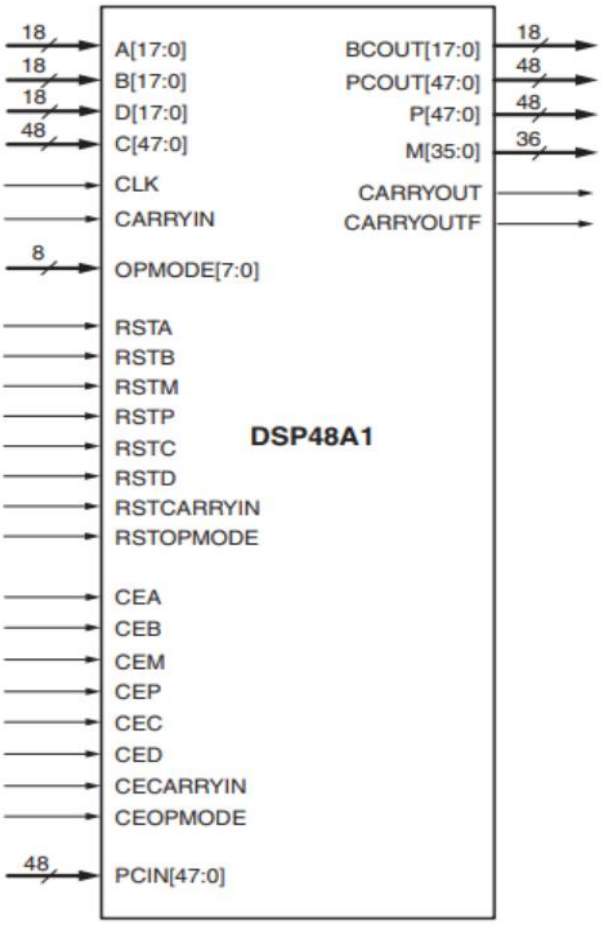
DSP48A1

BY:
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Abstract

The **DSP48A1** is a high-performance arithmetic block embedded within Xilinx Spartan-6 FPGA devices, designed to accelerate computation-intensive digital signal processing applications.

Its internal architecture is composed of three primary stages:

- A **pre-adder/subtractor** for operand conditioning
- An **18×18-bit multiplier** for high-speed arithmetic operations
- A **48-bit post-adder/subtractor** for accumulation and final computation

The functionality of the DSP block is dynamically controlled via an **8-bit OPMODE signal**, which governs internal multiplexers and enables flexible arithmetic operations such as **multiply-accumulate (MAC)** and other DSP functions.

To enhance performance and timing efficiency, the design incorporates **configurable pipeline registers** (A, B, C, D, M, and P), allowing optimization of latency and throughput. Additionally, **dedicated cascade ports** are utilized to chain multiple DSP blocks together, enabling scalable implementation of wider and more complex arithmetic operations.

The complete design flow includes:

- **Hardware Description:** Implemented in **Verilog HDL**
- **Simulation & Verification:** Conducted using QuestaSim
- **Synthesis & Implementation:** Performed using Xilinx Vivado

Design:-

```
module SyncRegMux #(
    parameter W      = 18,
    parameter ENABLE = 1,
    parameter R_MODE = "SYNC"
)(
    input  [W-1:0] data_in,
    input          clk, en, reset,
    output reg [W-1:0] data_out
);
    generate
        if (ENABLE) begin
            if (R_MODE == "SYNC") begin
                always @(posedge clk) begin
                    if (reset) data_out <= 0;
                    else if (en) data_out <= data_in;
                end
            end else begin
```

```
                always @(posedge clk or posedge reset) begin
                    if (reset) data_out <= 0;
                    else if (en) data_out <= data_in;
                end
            end
        end else begin
            always @(*) data_out = data_in;
        end
    endgenerate
endmodule


module Selector4to1 #(parameter BUS_W = 48) (
    input  [BUS_W-1:0] ch0, ch1, ch2, ch3,
    input  [1:0]       sel,
    output reg [BUS_W-1:0] out_data
);
    always @(*) begin
        case (sel)
            2'b00:   out_data = ch0;
            2'b01:   out_data = ch1;
            2'b10:   out_data = ch2;
            2'b11:   out_data = ch3;
            default: out_data = {BUS_W{1'b0}};
        endcase
    end
endmodule


module DSP48A1 (
    input  [17:0] A, B, D, BCIN,
    input  [47:0] C, PCIN,
    input  [7:0]  OPMODE,
    input         clk, CARRYIN,
    input         RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPMODE,
    input         CEA, CEB, CEM, CEP, CEC, CED, CECARRYIN, CEOPMODE,
    output [17:0] BCOUT,
    output [47:0] PCOUT, P,
    output [35:0] M,
    output        CARRYOUT, CARRYOUTF
);
    parameter A0REG=0, A1REG=1, B0REG=0, B1REG=1, CREG=1, DREG=1, MREG=1, PREG=1;
    parameter CARRYINREG=1, CARRYOUTREG=1, OPMODEREG=1;
    parameter CARRYINSEL="OPMODE5", B_INPUT="DIRECT", RSTTYPE="SYNC";
```

```
    wire [17:0] b_input_wire, b0_q, d_q, a0_q, a1_q, pre_adder_res, b1_mux_out,
b1_q;
    wire [35:0] multiplier_res, m_reg_q;
    wire [47:0] c_reg_q, x_mux_q, z_mux_q, post_adder_res, bus_concat;
    wire [7:0]  opmode_q;
    wire        cin_mux_q, cin_reg_q, cout_res;

    assign b_input_wire = (B_INPUT == "DIRECT") ? B : BCIN;

    SyncRegMux #(18, DREG,     RSTTYPE) REG_D  (D,           clk,
CED,      RSTD,      d_q     );
    SyncRegMux #(18, B0REG,    RSTTYPE) REG_B0 (b_input_wire, clk,
CEB,      RSTB,      b0_q    );
    SyncRegMux #(18, A0REG,    RSTTYPE) REG_A0 (A,           clk,
CEA,      RSTA,      a0_q    );
    SyncRegMux #(48, CREG,     RSTTYPE) REG_C  (C,           clk,
CEC,      RSTC,      c_reg_q );
    SyncRegMux #(8,  OPMODEREG,RSTTYPE) REG_OP (OPMODE,      clk, CEOPMODE,
RSTOPMODE, opmode_q);

    assign pre_adder_res = opmode_q[6] ? (d_q - b0_q) : (d_q + b0_q);
    assign b1_mux_out    = opmode_q[4] ? pre_adder_res : b0_q;

    SyncRegMux #(18, B1REG, RSTTYPE) REG_B1 (b1_mux_out, clk, CEB, RSTB, b1_q);
    assign BCOUT = b1_q;

    SyncRegMux #(18, A1REG, RSTTYPE) REG_A1 (a0_q, clk, CEA, RSTA, a1_q);

    assign multiplier_res = a1_q * b1_q;
    SyncRegMux #(36, MREG, RSTTYPE) REG_M (multiplier_res, clk, CEM, RSTM,
m_reg_q);
    assign M = m_reg_q;

    assign bus_concat = {d_q[11:0], a1_q, b1_q};

    Selector4to1 #(48) MUX_X (48'd0, {12'd0, m_reg_q}, P, bus_concat,
opmode_q[1:0], x_mux_q);
    Selector4to1 #(48) MUX_Z (48'd0, PCIN, P, c_reg_q, opmode_q[3:2], z_mux_q);

    assign cin_mux_q = (CARRYINSEL == "OPMODE5") ? opmode_q[5] :
                       (CARRYINSEL == "CARRYIN")  ? CARRYIN     : 1'b0;

    SyncRegMux #(1, CARRYINREG, RSTTYPE) REG_CIN (cin_mux_q, clk, CECARRYIN,
RSTCARRYIN, cin_reg_q);
```

```
    // FIX: extend to 49-bit to capture carry/borrow correctly
    assign {cout_res, post_adder_res} =
        opmode_q[7] ? ({1'b0, z_mux_q} - ({1'b0, x_mux_q} + cin_reg_q)) :
                      ({1'b0, z_mux_q} + {1'b0, x_mux_q} + cin_reg_q);

    SyncRegMux #(48, PREG,        RSTTYPE) REG_P    (post_adder_res, clk,
CEP,       RSTP,       P       );
    SyncRegMux #(1,  CARRYOUTREG, RSTTYPE) REG_COUT (cout_res,       clk,
CECARRYIN, RSTCARRYIN, CARRYOUT);
    assign PCOUT     = P;
    assign CARRYOUTF = CARRYOUT;

endmodule
```

TB:-

```
module DSP48A1_Verification_Env();

    reg [17:0] a_in, b_in, d_in, bcin_in;
    reg [47:0] c_in, pcin_in;
    reg [7:0]  op_ctrl;
    reg        sys_clk, cin_val;

    reg rsta, rstb, rstm, rstp, rstc, rstd, rstcin, rstop;
    reg cea, ceb, cem, cep, cec, ced, cecin, ceop;

    wire [17:0] bcout_out;
    wire [47:0] pcout_out, p_out;
    wire [35:0] m_out;
    wire        cout_out, coutf_out;

    // DUT instantiation
    DSP48A1 DUT (
        .A        (a_in),      .B       (b_in),     .D      (d_in),
        .C        (c_in),      .clk     (sys_clk),  .CARRYIN(cin_val),
        .OPMODE   (op_ctrl),   .BCIN    (bcin_in),
        .RSTA     (rsta),      .RSTB    (rstb),      .RSTM   (rstm),
        .RSTP     (rstp),      .RSTC    (rstc),      .RSTD   (rstd),
        .RSTCARRYIN(rstcin),   .RSTOPMODE(rstop),
        .CEA      (cea),       .CEB     (ceb),       .CEM    (cem),
        .CEP      (cep),       .CEC     (cec),       .CED    (ced),
        .CECARRYIN(cecin),     .CEOPMODE(ceop),
        .PCIN     (pcin_in),
        .BCOUT    (bcout_out), .PCOUT   (pcout_out), .P      (p_out),
```

```
    .M        (m_out),     .CARRYOUT(cout_out),  .CARRYOUTF(coutf_out)
);

// Clock generation: 10 ns period
always #5 sys_clk = ~sys_clk;

initial begin
    sys_clk = 0;
    reset_system();

    // Test Scenario 1: Basic Operation
    run_test(8'b11011101, 20, 10, 25, 350);
    #40;

    $display("Simulation Finished Successfully");
    $finish;
end

task reset_system();
    begin
        // Initialise all data/control inputs to avoid X-propagation
        a_in    = 18'd0;
        b_in    = 18'd0;
        d_in    = 18'd0;
        bcin_in = 18'd0;   // FIX: was never assigned
        c_in    = 48'd0;
        pcin_in = 48'd0;   // FIX: was never assigned
        op_ctrl = 8'd0;
        cin_val = 1'b0;    // FIX: was never assigned

        {rsta, rstb, rstm, rstp, rstc, rstd, rstcin, rstop} = 8'hFF;
        {cea,  ceb,  cem,  cep,  cec,  ced,  cecin,  ceop}  = 8'hFF;

        repeat(2) @(negedge sys_clk);

        {rsta, rstb, rstm, rstp, rstc, rstd, rstcin, rstop} = 8'h00;
    end
endtask

task run_test(
    input [7:0]  op,
    input [17:0] a, b, d,
    input [47:0] c
);
    begin
```

```
            op_ctrl = op;
            a_in    = a;
            b_in    = b;
            d_in    = d;
            c_in    = c;
            @(negedge sys_clk);
        end
    endtask

endmodule
```

##5. Lint Control File (Verilator / Generic Lint Checker)

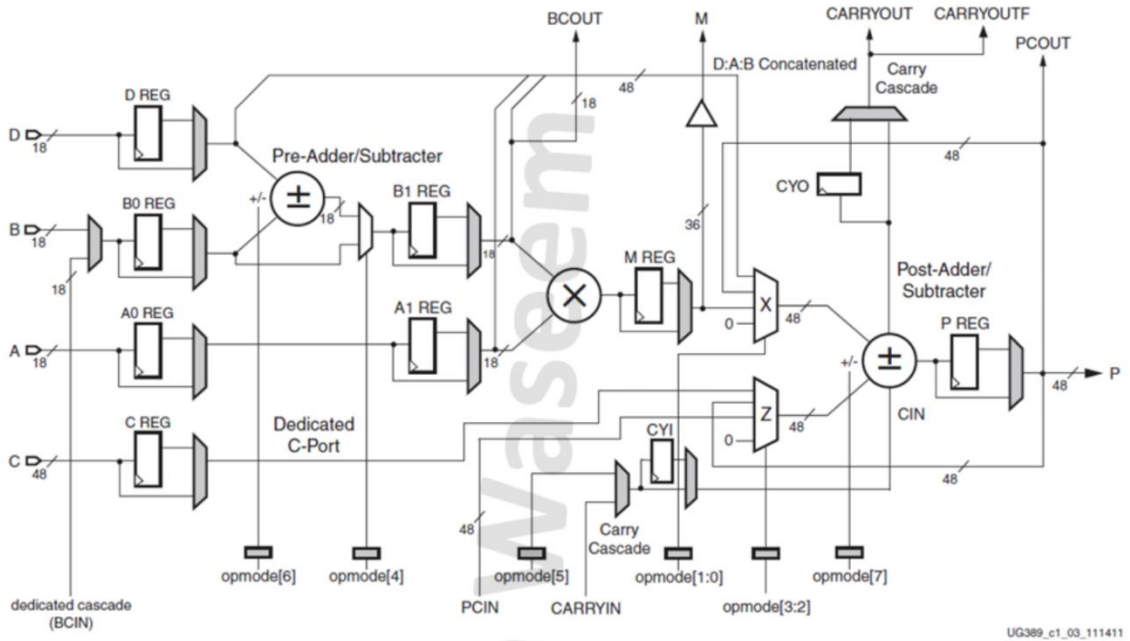
// Lint Control File for DSP48A1
// Tool: Verilator / Generic Lint Checker

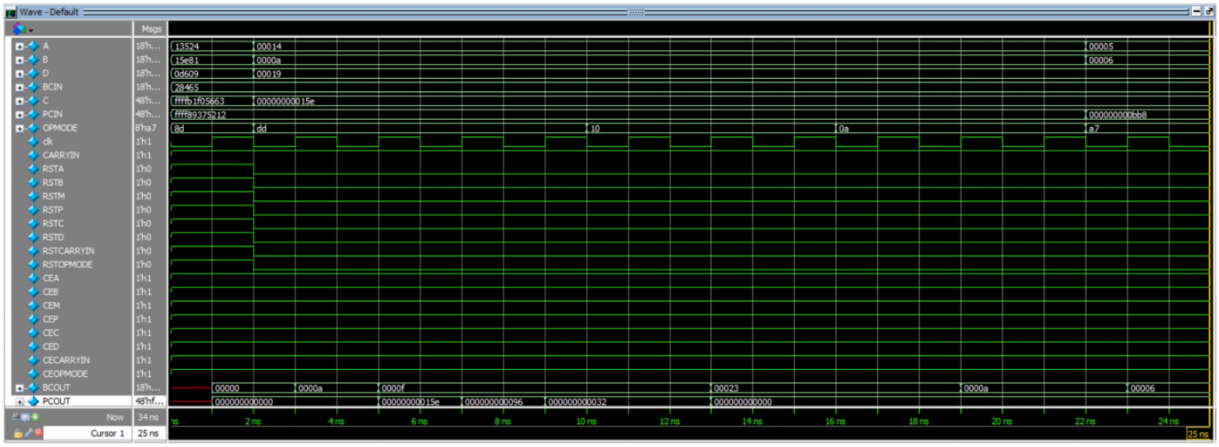
// Rule: Ignore bit width mismatch in multiplier because it's intentional (18x18 = 36)
lint_off -rule WIDTH -file "DSP48A1_Redesign.v" -lines 60-70

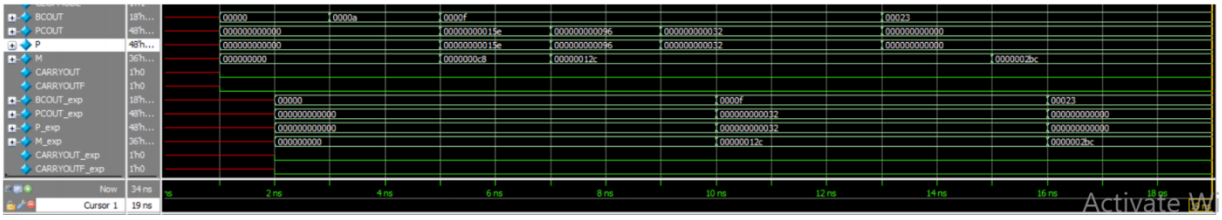
// Rule: Allow unused bits in concatenation if specified by architecture
lint_off -rule UNUSED -file "DSP48A1_Redesign.v" -match "bus_concat*"

// Rule: Verify Synchronous Reset Style matches parameter
lint_on -rule SYNC_RESET

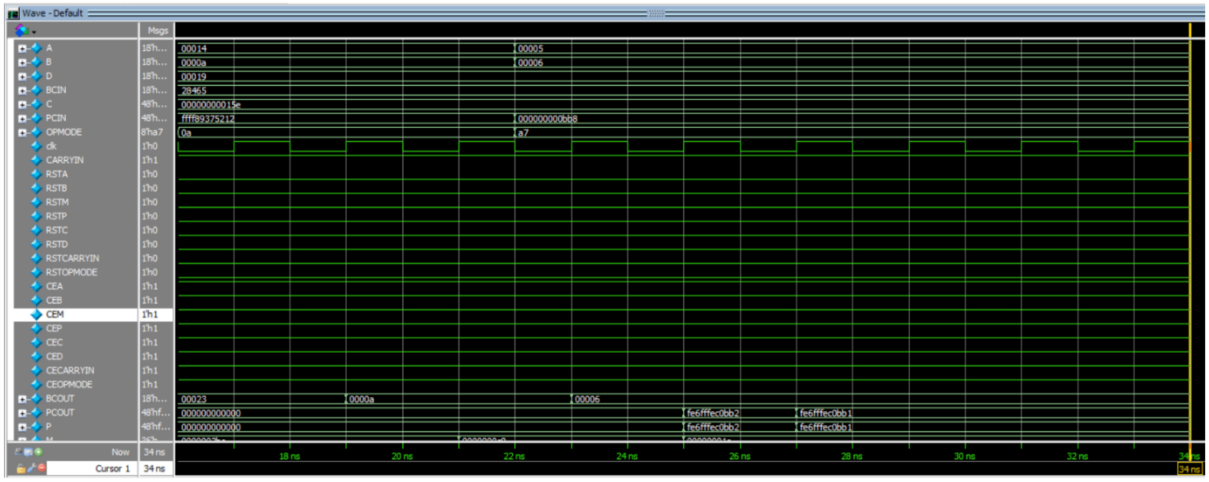
// Rule: Check for potential combinational loops in Muxes
lint_on -rule COMBLOOP
```
```

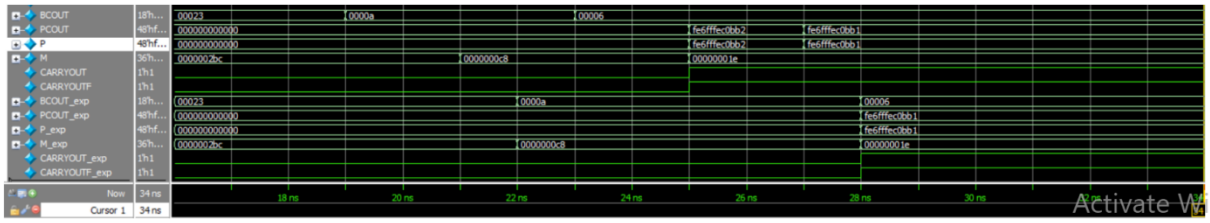






Activate Wir


Wave - Default
Msgs
A
B
D
BCIN
C
PCIN
OPMODE
clk
CARRYIN
RSTA
RSTB
RSTM
RSTP
RSTC
RSTD
RSTCARRYIN
RSTOPMODE
CEA
CEB
CEM
CEP
CEC
CED
CECARRYIN
CEOPMODE
BCOUT
PCOUT
P
Now
Cursor 1
18 ns
20 ns
22 ns
24 ns
26 ns
28 ns
30 ns
32 ns
34 ns


BCOUT
PCOUT
P
M
CARRYOUT
CARRYOUTF
BCOUT_exp
PCOUT_exp
P_exp
M_exp
CARRYOUT_exp
CARRYOUTF_exp
Now
Cursor 1
18 ns
20 ns
22 ns
24 ns
26 ns
28 ns
30 ns

Constraint File:-

File Edit View

```
## This file is a general .xdc for the Basys3 rev B board
## To use it in a project:
## - uncomment the lines corresponding to used pins
## - rename the used ports (in each line, after get_ports) according to the top level signal names in the project

## Clock signal
set_property -dict {PACKAGE_PIN W5 IOSTANDARD LVCMOS33} [get_ports clk]
create_clock -period 10.000 -name sys_clk_pin -waveform {0.000 5.000} [get_ports clk]

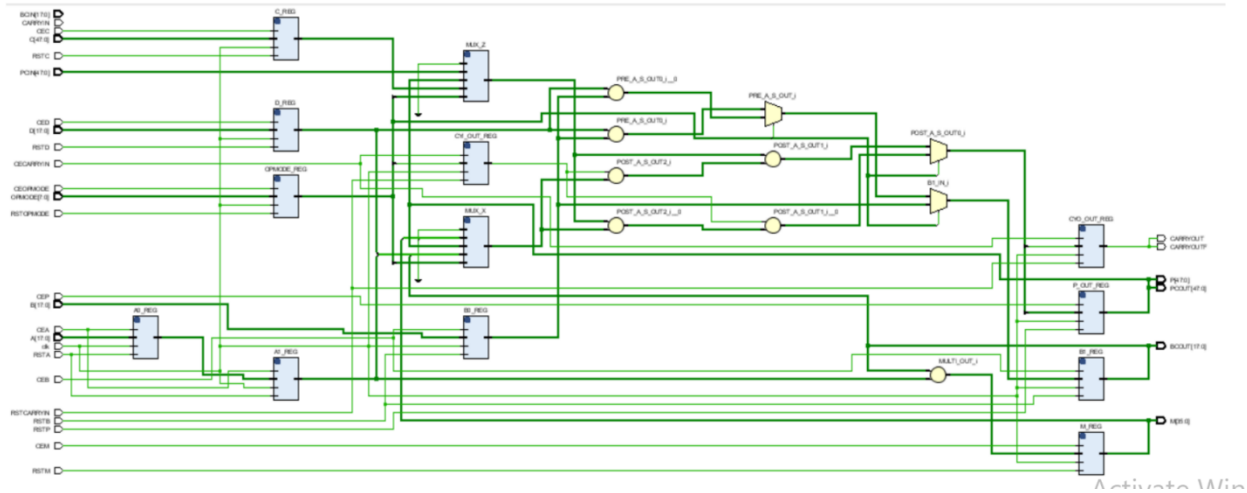

## Switches
#set_property -dict { PACKAGE_PIN V17   IOSTANDARD LVCMOS33 } [get_ports {sw[0]}]
#set_property -dict { PACKAGE_PIN V16   IOSTANDARD LVCMOS33 } [get_ports {sw[1]}]
#set_property -dict { PACKAGE_PIN W16   IOSTANDARD LVCMOS33 } [get_ports {sw[2]}]
#set_property -dict { PACKAGE_PIN W17   IOSTANDARD LVCMOS33 } [get_ports {sw[3]}]
#set_property -dict { PACKAGE_PIN W15   IOSTANDARD LVCMOS33 } [get_ports {sw[4]}]
#set_property -dict { PACKAGE_PIN V15   IOSTANDARD LVCMOS33 } [get_ports {sw[5]}]
#set_property -dict { PACKAGE_PIN W14   IOSTANDARD LVCMOS33 } [get_ports {sw[6]}]
#set_property -dict { PACKAGE_PIN W13   IOSTANDARD LVCMOS33 } [get_ports {sw[7]}]
#set_property -dict { PACKAGE_PIN V2    IOSTANDARD LVCMOS33 } [get_ports {sw[8]}]
#set_property -dict { PACKAGE_PIN T3    IOSTANDARD LVCMOS33 } [get_ports {sw[9]}]
#set_property -dict { PACKAGE_PIN T2    IOSTANDARD LVCMOS33 } [get_ports {sw[10]}]
#set_property -dict { PACKAGE_PIN R3    IOSTANDARD LVCMOS33 } [get_ports {sw[11]}]
#set_property -dict { PACKAGE_PIN W2    IOSTANDARD LVCMOS33 } [get_ports {sw[12]}]
#set_property -dict { PACKAGE_PIN U1    IOSTANDARD LVCMOS33 } [get_ports {sw[13]}]
#set_property -dict { PACKAGE_PIN T1    IOSTANDARD LVCMOS33 } [get_ports {sw[14]}]
#set_property -dict { PACKAGE_PIN R2    IOSTANDARD LVCMOS33 } [get_ports {sw[15]}]


## LEDs
#set_property -dict { PACKAGE_PIN U16   IOSTANDARD LVCMOS33 } [get_ports {led[0]}]
#set_property -dict { PACKAGE_PIN E19   IOSTANDARD LVCMOS33 } [get_ports {led[1]}]
#set_property -dict { PACKAGE_PIN U19   IOSTANDARD LVCMOS33 } [get_ports {led[2]}]
#set_property -dict { PACKAGE_PIN V19   IOSTANDARD LVCMOS33 } [get_ports {led[3]}]
#set_property -dict { PACKAGE_PIN W18   IOSTANDARD LVCMOS33 } [get_ports {led[4]}]
#set_property -dict { PACKAGE_PIN U15   IOSTANDARD LVCMOS33 } [get_ports {led[5]}]
#set_property -dict { PACKAGE_PIN U14   IOSTANDARD LVCMOS33 } [get_ports {led[6]}]
#set_property -dict { PACKAGE_PIN V14   IOSTANDARD LVCMOS33 } [get_ports {led[7]}]
#set_property -dict { PACKAGE_PIN V13   IOSTANDARD LVCMOS33 } [get_ports {led[8]}]
#set_property -dict { PACKAGE_PIN V3    IOSTANDARD LVCMOS33 } [get_ports {led[9]}]
```

Elaboration Messages:-



Schematic Snippest



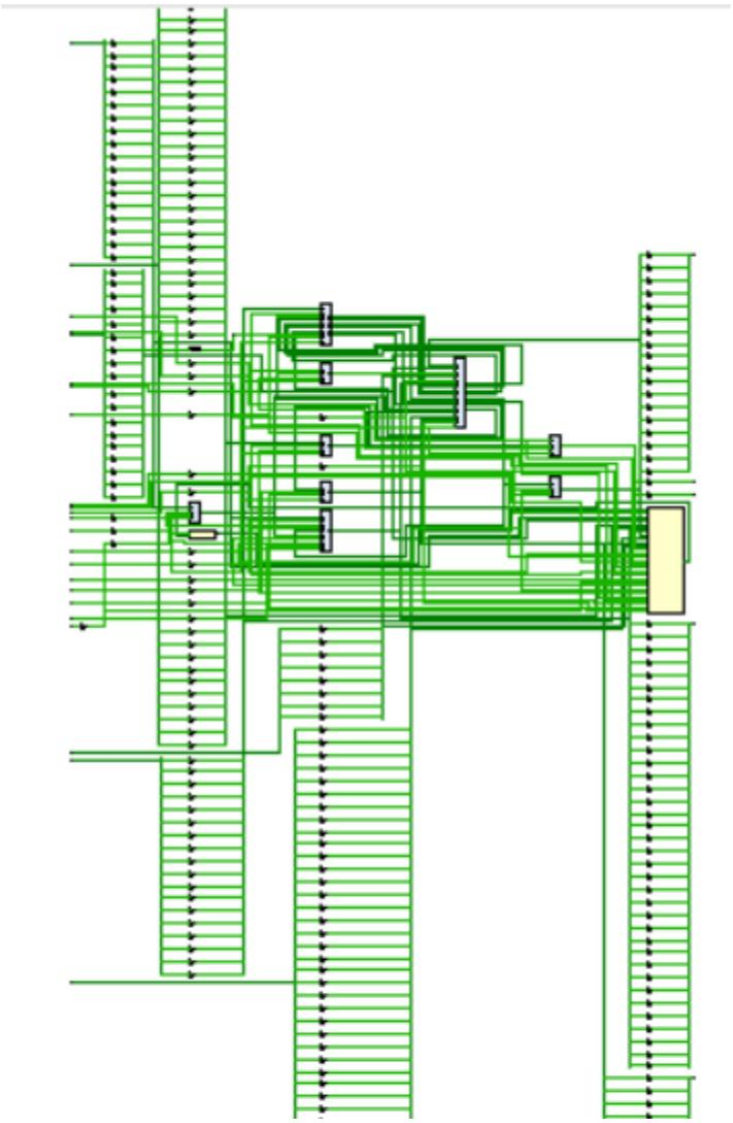
Timing :-

Design Timing Summary

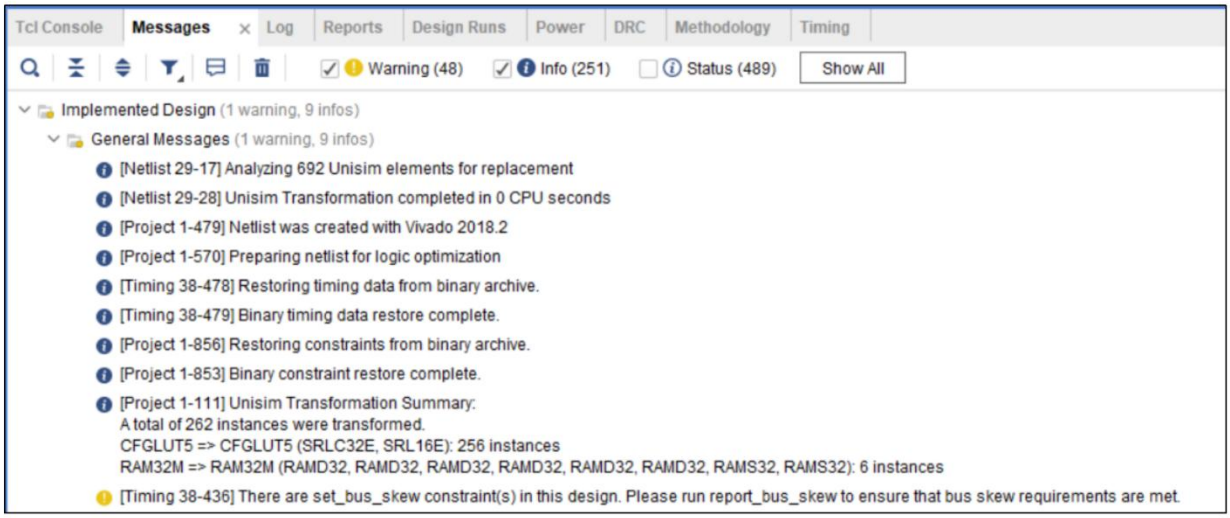
Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	5.164 ns	Worst Hold Slack (WHS):	0.182 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	106	Total Number of Endpoints:	106	Total Number of Endpoints:	162

All user specified timing constraints are met.

Schematic Snippets:-



Implementation:-



Utilization:-

Name	Slice LUTs (133800)	Slice Registers (267600)	F7 Muxes (66900)	F8 Muxes (33450)	Slice (33450)	LUT as Logic (133800)	LUT as Memory (46200)	LUT Flip Flop Pairs (133800)	Block RAM Tile (365)	DSPs (740)	Bonded IOB (500)	BUFGCTRL (32)
DSP48A1	2703	4225	96	12	1443	2229	474	1565	8	1	327	2
A1_REG (DFF_and_m...	0	18	0	0	7	0	0	0	0	0	0	0
B1_REG (DFF_and_m...	0	18	0	0	9	0	0	0	0	0	0	0
C_REG (DFF_and_mu...	0	48	0	0	18	0	0	0	0	0	0	0
CYI_OUT_REG (DFF_...	1	1	0	0	1	1	0	1	0	0	0	0
CYO_OUT_REG (DFF...	0	1	0	0	1	0	0	0	0	0	0	0
D_REG (DFF_and_mu...	0	18	0	0	10	0	0	0	0	0	0	0
dbg_hub (dbg_hub)	475	727	0	0	246	451	24	303	0	0	0	1
M_REG (DFF_and_mu...	0	0	0	0	0	0	0	0	0	1	0	0
OPMODE_REG (DFF_...	228	8	0	0	70	228	0	0	0	0	0	0
P_OUT_REG (DFF_an...	0	48	0	0	12	0	0	0	0	0	0	0
u_ila_0 (u_ila_0)	1998	3338	96	12	1129	1548	450	1209	8	0	0	0

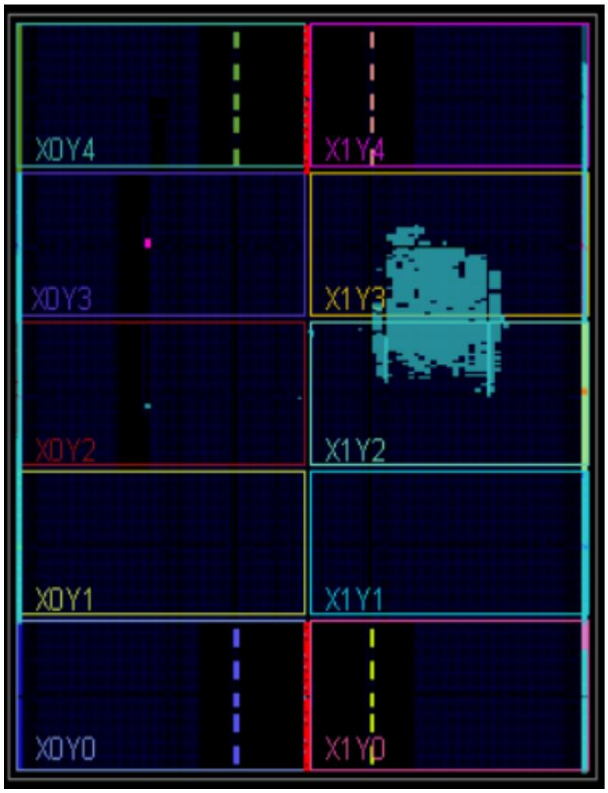
Timing:-

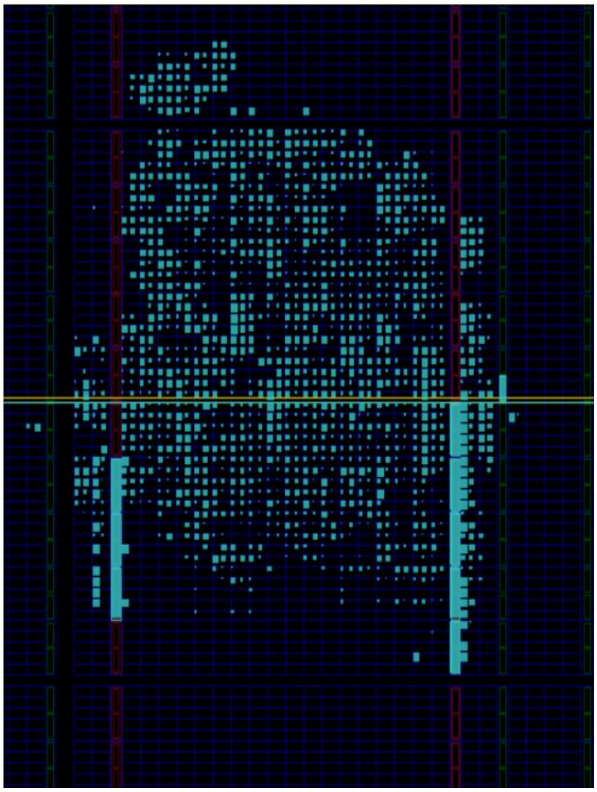
Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	3.384 ns	Worst Hold Slack (WHS):	0.050 ns	Worst Pulse Width Slack (WPWS):	3.950 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	8077	Total Number of Endpoints:	8061	Total Number of Endpoints:	5136

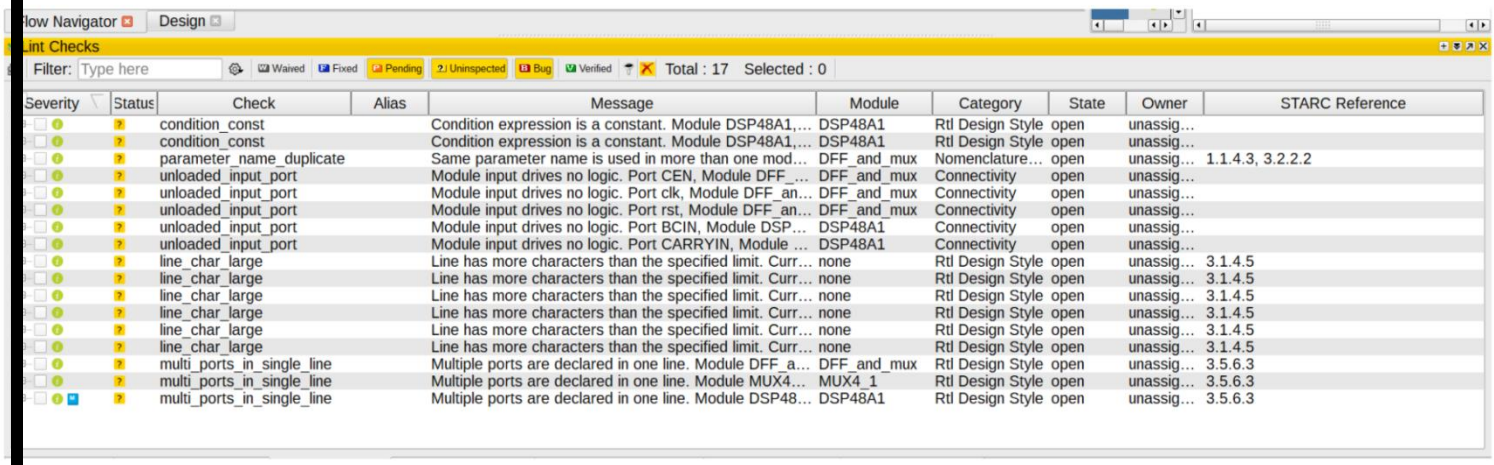
All user specified timing constraints are met.

Device snippet:-

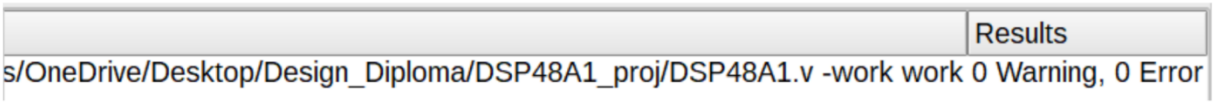


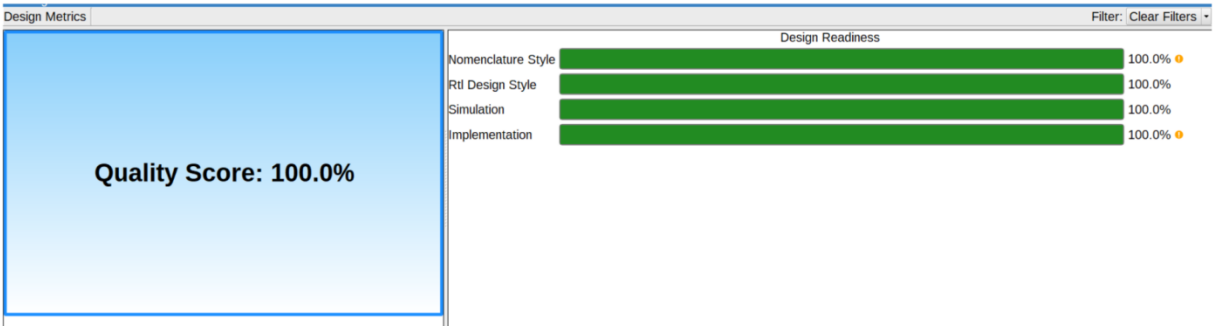


Linting :-

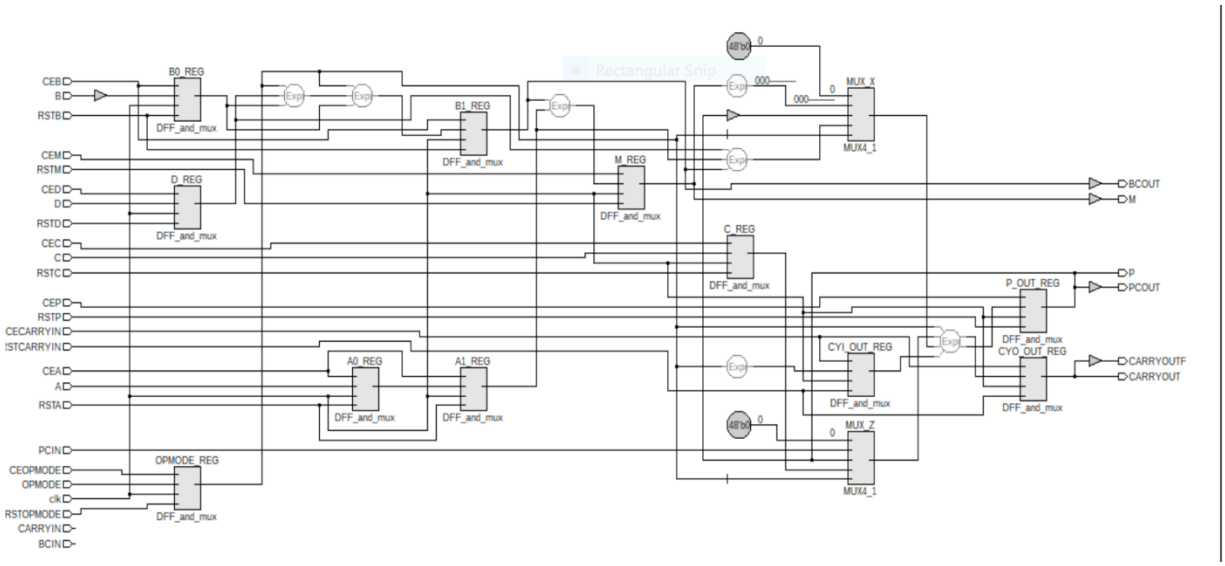


Severity	Status	Check	Alias	Message	Module	Category	State	Owner	STARC Reference
		condition_const		Condition expression is a constant. Module DSP48A1,...	DSP48A1	Rtl Design Style	open	unassig...	
		condition_const		Condition expression is a constant. Module DSP48A1,...	DSP48A1	Rtl Design Style	open	unassig...	
		parameter_name_duplicate		Same parameter name is used in more than one mod...	DFF_and_mux	Nomenclature...	open	unassig...	1.1.4.3, 3.2.2.2
		unloaded_input_port		Module input drives no logic. Port CEN, Module DFF_...	DFF_and_mux	Connectivity	open	unassig...	
		unloaded_input_port		Module input drives no logic. Port clk, Module DFF_an...	DFF_and_mux	Connectivity	open	unassig...	
		unloaded_input_port		Module input drives no logic. Port rst, Module DFF_an...	DFF_and_mux	Connectivity	open	unassig...	
		unloaded_input_port		Module input drives no logic. Port BCIN, Module DSP...	DSP48A1	Connectivity	open	unassig...	
		unloaded_input_port		Module input drives no logic. Port CARRYIN, Module ...	DSP48A1	Connectivity	open	unassig...	
		line_char_large		Line has more characters than the specified limit. Curr...	none	Rtl Design Style	open	unassig...	3.1.4.5
		line_char_large		Line has more characters than the specified limit. Curr...	none	Rtl Design Style	open	unassig...	3.1.4.5
		line_char_large		Line has more characters than the specified limit. Curr...	none	Rtl Design Style	open	unassig...	3.1.4.5
		line_char_large		Line has more characters than the specified limit. Curr...	none	Rtl Design Style	open	unassig...	3.1.4.5
		line_char_large		Line has more characters than the specified limit. Curr...	none	Rtl Design Style	open	unassig...	3.1.4.5
		line_char_large		Line has more characters than the specified limit. Curr...	none	Rtl Design Style	open	unassig...	3.1.4.5
		multi_ports_in_single_line		Multiple ports are declared in one line. Module DFF_a...	DFF_and_mux	Rtl Design Style	open	unassig...	3.5.6.3
		multi_ports_in_single_line		Multiple ports are declared in one line. Module MUX4...	MUX4_1	Rtl Design Style	open	unassig...	3.5.6.3
		multi_ports_in_single_line		Multiple ports are declared in one line. Module DSP48...	DSP48A1	Rtl Design Style	open	unassig...	3.5.6.3





Schematic by Questlint :-


CEB
B
RSTB
B0_REG
DFF_and_mux
B1_REG
DFF_and_mux
CEM
RSTM
CED
D
RSTD
D_REG
DFF_and_mux
M_REG
DFF_and_mux
CEC
C
RSTC
C_REG
DFF_and_mux
CEP
RSTP
CECARRYIN
!STCARRYIN
CEA
A
RSTA
A0_REG
DFF_and_mux
A1_REG
DFF_and_mux
PCIN
CEOPMODE
OPMODE
clk
RSTOPMODE
CARRYIN
BCIN
OPMODE_REG
DFF_and_mux
48'b0
0
000
000
MUX_X
MUX4_1
BCOUT
M
P_OUT_REG
DFF_and_mux
P
PCOUT
CYI_OUT_REG
DFF_and_mux
CYO_OUT_REG
DFF_and_mux
CARRYOUTF
CARRYOUT
48'b0
0
MUX_Z
MUX4_1